UNDERSTANDING OUTCOME-BASED EDUCATION (OBE)

Prepared for :

Students,
Dept of CSE









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Version 1.0

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Learning Objectives

After this lecture, students will be able to:

- Explain the meaning of Outcome-Based Education (OBE)
- Understand the difference between traditional education and OBE
- Explain Course Outcomes (COs) and Program Outcomes (POs)
- Understand Bloom's Taxonomy
- Identify different levels of learning
- Understand how OBE questions are designed
- Understand how OBE affects quizzes, exams, assignments, labs, and projects
- Prepare effectively for OBE-based assessments

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Outcome-Based Education

Outcome-Based Education (OBE) is an educational approach that focuses on what students are expected to **achieve at the end of a program or course**. The emphasis is placed on learning outcomes along with teaching content. In OBE, success is measured by students' ability to demonstrate knowledge, skills, and professional attitudes.

Key Principles of OBE

OBE is based on several core principles:

- Clearly defined learning outcomes
- Alignment between course outcomes, teaching content, and assessment plan
- Student-centered learning
- Evidence-based assessment
- Continuous Quality Improvement (CQI)

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These principles ensure that learning activities and assessments are directly linked to expected course outcomes.

OBE in Engineering Education

In engineering education, OBE ensures that graduates are capable of solving complex engineering problems and can function effectively in professional environments. Board of Accreditation for Engineering & Technical Education (BAETE) adopts OBE to ensure that engineering programs produce graduates with appropriate technical competence, ethical responsibility, and societal awareness.

Role of Outcomes in OBE

OBE operates through a hierarchy of outcomes:

- **Program Educational Objectives (PEOs)** describe graduates' achievements after a few years of graduation.

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- **Program Outcomes (POs)** define competencies students must have at the time of graduation.
- **Course Outcomes (COs)** specify what students achieve in individual courses.

This structured outcome framework ensures coherence across the entire program.

Assessment and Continuous Quality Improvement

Assessment is an essential component of OBE. Both direct and indirect assessment methods are used to measure outcome attainment. The results are analyzed to identify gaps and implement corrective actions. This continuous feedback process forms the basis of **Continuous Quality Improvement**, a mandatory requirement of BAETE.

Importance of OBE for Accreditation

Accreditation is not merely a formal recognition. It is a public assurance that an academic program meets established standards of quality and prepares graduates for professional practice. In this context, Outcome-Based Education (OBE) plays a central role. Modern accreditation bodies evaluate programs not only on infrastructure, curriculum structure, or faculty qualifications, but on clear evidence that students are achieving defined learning outcomes.

OBE provides the structure that accreditation requires. It ensures that Vision and Mission statements are translated into measurable Program Educational Objectives (PEOs), that Program Outcomes (POs) reflect expected graduate attributes, and that Course Outcomes (COs) are carefully aligned to those program-level goals. This alignment demonstrates that learning is intentional, systematic, and purposeful rather than incidental.

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Accreditation agencies such as the Board of Accreditation for Engineering and Technical Education (BAETE) emphasize measurable attainment, constructive alignment, and Continuous Quality Improvement (CQI). Without OBE, it becomes difficult to show how teaching activities lead to outcome achievement, how assessments measure intended competencies, or how collected data is used to improve the program. OBE makes these connections visible and defensible.

Another critical aspect is evidence. Accreditation is evidence-driven. Programs must present documented proof of outcome attainment, assessment methods, attainment calculations, stakeholder feedback, and improvement actions. OBE provides the framework through which this evidence is generated and organized. When implemented properly, it transforms accreditation preparation from a last-minute documentation exercise into an ongoing, structured academic practice.

Ultimately, OBE strengthens not only accreditation compliance but also academic integrity. It ensures that graduates possess the knowledge, skills, and professional attributes promised by the program. In doing so, it builds trust among students, employers, regulatory bodies, and society at large.

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Vision, Mission & Program Educational Objectives

Program Vision and Mission define the long-term direction and purpose of an academic program. The Vision reflects the aspirations of the program, highlighting what it aims to achieve in the future.

The mission describes how the program intends to fulfill that vision through quality education, research, innovation, and service. Together, they provide a strategic foundation that guides the development of PEOs, POs, and overall academic planning.

Program Educational Objectives, commonly known as PEOs, describe what graduates are expected to achieve a few years after completing a program. Unlike course outcomes or program outcomes, PEOs do not focus on what students learn inside the classroom. Instead, they focus on what graduates become in their professional life.

In simple terms, PEOs answer this question:

"What should our graduates be doing 3–5 years after graduation?"

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PEOs provide a long-term vision for an academic program and guide curriculum design, teaching strategies, assessment, and continuous improvement.

2.2 Purpose of PEOs in Outcome Based Education

PEOs play a strategic role in Outcome Based Education. They act as a bridge between the institution's vision and the actual impact of the program on graduates and society.

The main purposes of PEOs are to:

- Align the program with the program's mission and vision
- Reflect the needs of industry, society, and stakeholders
- Guide the formulation of Program Outcomes (POs)
- Support accreditation and quality assurance requirements
- Ensure graduates remain competent in a challenges professional environment

Without clearly defined PEOs, OBE becomes fragmented and loses its long-term direction.

Vision of Department of CSE

"Be the forerunner towards a global knowledge hub with striving for the pursuit of excellence in CSE education, research, entrepreneurship, and technological services to the society."

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Mission of Department of CSE

The Department of CSE is committed to:

⇒ *Achieve academic and research excellence through innovative ICT-based teaching*
⇒ *Prepare students for challenges in the ICT industry*
⇒ *Promote research and collaboration*
⇒ *Ensure quality and ethical values*
⇒ *Develop entrepreneurial and leadership skills*

Program Educational Objectives (PEOs) of CSE Dept.

PEO1: *Apply CSE knowledge and modern tools to solve real-life problems.*

PEO2: *Design, develop, and maintain computing systems.*

PEO3: *Communicate effectively and lead teams.*

PEO4: *Evaluate systems considering quality, security, ethics, and cost.*

PEO5: *Engage in lifelong learning and adapt to change.*

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Mapping of PEO with Mission

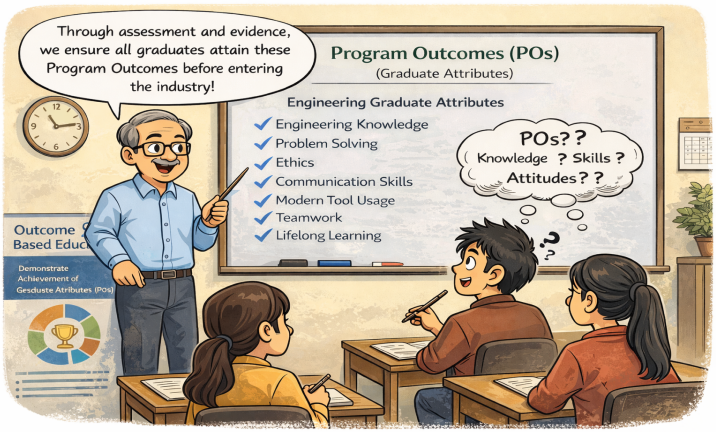
PEOs No.	M1	M2	M3	M4	M5
PEO1	✓			✓	
PEO2	✓	✓			
PEO3			✓		✓
PEO4	✓				✓
PEO5				✓	✓

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Program Outcomes (POs)

Program Outcomes (POs), also known as **graduate attributes**, are narrower statements that describe what students are expected to **know** and **be able to do by the time of graduation**. These outcomes focus on the knowledge, skills, and attitudes acquired by students throughout their academic journey in the program.

In Outcome Based Education, Program Outcomes ensure that graduates achieve an **acceptable minimum level of competence** before entering professional practice or higher studies. Every academic program must demonstrate, through assessment and evidence, that these outcomes are being achieved.



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BAETE specifically requires that engineering graduates acquire a defined set of graduate attributes. These attributes form the foundation of Program Outcomes for all accredited engineering programs.

PO No.	Title	Description
PO1	**Engineering Knowledge**	Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	**Problem Analysis**	Identify, formulate, review research literature, and analyze complex engineering problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	**Design and Development of Solutions**	Design solutions for complex engineering problems and design systems, components, or processes that meet specified needs with appropriate consideration for public health, safety, cultural, societal, and environmental factors.
PO4	**Conduct Investigations of Complex Problems**	Use research-based knowledge and methods, including design of experiments, data analysis, and interpretation, to provide valid conclusions.
PO5	**Modern Tool Usage**	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, with an understanding of limitations.
PO6	**The Engineer and Society**	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues and responsibilities relevant to engineering practice.
PO7	**Environment and Sustainability**	Understand the impact of engineering solutions in societal and environmental contexts and demonstrate knowledge of sustainable development.
PO8	**Ethics**	Apply ethical principles and commit to professional ethics, responsibilities, and norms of engineering practice.
PO9	**Individual and Team Work**	Function effectively as an individual and as a member or leader in diverse and multidisciplinary teams.
PO10	**Communication**	Communicate effectively on complex engineering activities with the engineering community and society, including writing reports, making presentations, and giving/receiving instructions.
PO11	**Project Management and Finance**	Demonstrate knowledge of engineering and management principles and apply them to manage projects as a team member or leader in multidisciplinary environments.

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PO12	**Life-Long Learning**	Recognize the need for and have the ability to engage in independent, life-long learning in the context of technological change.

Mapping with PEOs and POs

PEO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
PEO1	✓	✓			✓							
PEO2			✓		✓						✓	
PEO3								✓	✓	✓		
PEO4					✓	✓	✓					
PEO5												✓

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Course Outcomes (COs) and CO–PO Mapping

Course Outcomes (COs) describe what students are expected to **know and be able to do at the end of a course**. COs translate Program Outcomes into course-level achievements and guide teaching, learning activities, and assessment.

In Outcome-Based Education, every course must have clearly defined COs that contribute directly to one or more Program Outcomes.

Note: Many experts suggest mapping one PO to each CO, as this simplifies outcome attainment analysis and makes evaluation, documentation, and continuous quality improvement more manageable and transparent.

Importance of Course Outcomes in OBE

Course Outcomes play a central role in OBE implementation. They ensure that each course contributes meaningfully to the overall program.

The importance of COs includes:

- Providing clear learning targets for students
- Guiding assessment design and evaluation
- Supporting measurement of PO attainment
- Ensuring alignment between curriculum and outcomes

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Clearly defined Course Outcomes (COs) guide faculty in shifting from traditional content-driven instruction to a focused, outcome-oriented teaching approach where student achievement becomes the central priority.

Example:

Consider a course titled **Database Management Systems**. At the end of this course students are able to-

- **CO1: *Design*** a relational database schema for a given real-world problem using entity–relationship modeling techniques by the end of the course.

  (Specific: focuses on database design; Measurable: assessed through assignments or exams; Achievable: appropriate for the course scope; Relevant: aligned with problem analysis and design POs; Time-bound: by the end of the course.)

- **CO2: Apply** SQL commands to create, insert, update, and retrieve data from relational database tables.

  (Specific: SQL implementation; Measurable: lab performance; Achievable: within lab resources; Relevant: supports technical competency POs; Time-bound: completed during the course.)

- **CO3: *Analyze*** normalization problems and ***apply*** normalization techniques up to 3NF to reduce data redundancy in a given database case study.

  (Specific: normalization analysis; Measurable: quizzes or case studies; Achievable: fits course content; Relevant: supports analytical POs; Time-bound: achieved by course completion.)

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- **CO4: *Demonstrate*** ethical and professional responsibility in database usage by following data security and integrity guidelines in course projects.

  (Specific: ethical practice; Measurable: project evaluation; Achievable: through guided projects; Relevant: aligned with ethics-related POs; Time-bound: assessed at the end of the course.)

In this example, each Course Outcome is specific, measurable, and achievable within the course duration, relevant to program goals, and clearly time-bound. The course includes four well-focused COs, each written with clear action verbs that support objective assessment.

You can check your course outline to easily find the Course Outcomes (COs) for your courses.

CO Attainment Calculation

Course Outcome (CO) attainment is calculated through a structured and uniform assessment process. A common attainment threshold of 50% is applied across all courses. This threshold indicates the minimum level of performance required for a CO to be considered attained.

For each course, assessment items such as midterm examinations, final examinations, assignments, laboratory tasks, projects, presentations, and viva are mapped with specific COs. Each CO is measured separately based on the marks assigned to its corresponding assessment items.

The attainment percentage of an individual CO is calculated using the following formula:

CO_i (%) = (∑ Marks obtained for CO_i / ∑ Marks assigned for CO_i)×100

If the calculated percentage is 50% or above, the CO is considered attained by the student.

For example, if a student obtains 7 marks out of 13 in a question mapped to CO1:

CO1 (%) = (7 / 13) × 100 = 53.8%

Since 53.8% ≥ 50%, CO1 is considered attained by the student.

After calculating student-level CO attainment, course-level CO attainment is determined by calculating the percentage of students who achieved the specific CO.

Percentage of Students Achieving CO_i = (Number of students achieving CO_i / Total number of students) × 100

For example, if 9 out of 10 students attain CO1:

Percentage of Students Achieving CO1 = (9 / 10) × 100 = 90%

Since 90% ≥ 50%, CO1 is considered attained at the course level.

Finally, the attained CO values are used for PO attainment calculation through the approved CO–PO mapping matrix. The results are reviewed against the program benchmark, and necessary CQI actions are taken where performance gaps are identified.

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Bloom's Taxonomy in OBE

Bloom's Taxonomy was introduced in **1956** by **Benjamin Bloom** and his colleagues to classify educational objectives and support effective teaching and assessment. The original taxonomy organized cognitive learning into six levels, progressing from simple knowledge recall to higher-order thinking such as analysis and evaluation. In 2001, the framework was revised to use action-oriented verbs and to emphasize active learning, making it more suitable for Outcome-Based Education. Today, Bloom's Taxonomy is widely used to write measurable Course Outcomes, design assessments, and align teaching with expected learning outcomes.

Bloom's Taxonomy was originally developed for the cognitive domain, which focuses on intellectual skills such as remembering, understanding, applying, analyzing, evaluating, and creating. In most OBE practices, when we say "Bloom's level," we are usually referring to this cognitive hierarchy. However, Bloom and his colleagues also recognized two other domains:

Affective Domain – related to attitudes, values, ethics, and professional behavior.

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This includes levels such as receiving, responding, valuing, organizing, and internalizing values.

Psychomotor Domain – related to physical skills, technical execution, and hands-on performance.

This domain was later expanded by other researchers and includes levels such as imitation, manipulation, precision, articulation, and naturalization. So, Bloom's framework is not limited to cognitive learning. It covers cognitive, affective, and psychomotor domains.

That said, in engineering OBE practice, Bloom's taxonomy is most commonly applied to the cognitive domain when writing Course Outcomes and designing theory assessments. **Psychomotor levels** are especially relevant in **laboratory and project courses**, while **affective levels** are important for outcomes related to **ethics, teamwork, professionalism, and lifelong learning.** A complete OBE system ideally considers all three domains, depending on the nature of the course and the intended outcomes.

Bloom's Taxonomy is a framework used to classify learning objectives based on **levels of thinking**. In Outcome-Based Education, it helps faculty write **clear and measurable Course Outcomes**, design assessments, and ensure that students are not only memorizing content but also understanding, applying, and analyzing it.

Bloom's Taxonomy supports OBE by linking **what we expect students to achieve** with **how we teach and assess**.

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Bloom's Taxonomy Levels and Their Importance

Bloom's Taxonomy is commonly divided into six cognitive levels, arranged from lower to higher order thinking.

Importance of Bloom's level (Cognitive)

Bloom's Level	What Students Do	Importance in OBE
Remember	Recall facts, definitions, or concepts	Forms the foundation of learning and supports basic understanding
Understand	Explain ideas or concepts in own words	Shows that students truly grasp the subject matter
Apply	Use knowledge to solve problems	Demonstrates practical use of learned concepts
Analyze	Break problems into parts and examine relationships	Develops critical thinking and problem-solving skills
Evaluate	Make judgments based on criteria	Encourages decision-making and justification
Create	Design or develop new solutions	Represents highest level of learning and innovation

Importance of Bloom's Level (Psychomotor Domain)

Psychomotor Level	What Students Do	Importance in OBE
Imitation	Observe and replicate a demonstrated skill	Forms the starting point of technical skill development
Manipulation	Perform tasks with guidance or instructions	Shows ability to follow procedures correctly
Precision	Execute skills accurately and independently	Ensures technical competence and reliability
Articulation	Coordinate multiple skills smoothly	Demonstrates advanced technical integration
Naturalization	Perform skills automatically with high efficiency	Reflects mastery and professional-level performance

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Importance of Bloom's Level (Affective Domain)

Affective Level	What Students Do	Importance in OBE
Receiving	Show awareness and willingness to listen	Builds openness to learning and professional values
Responding	Participate actively and react to learning situations	Demonstrates engagement and involvement
Valuing	Show commitment to certain beliefs or standards	Reflects development of ethics and responsibility
Organizing	Integrate values into a consistent system	Helps students prioritize professional and social responsibilities
Characterizing	Consistently act according to internalized values	Indicates maturity, integrity, and professional behavior

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Knowledge Profiles

Knowledge Profiles describe the **types and depth of knowledge** that engineering students are expected to acquire during a program. In Outcome-Based Education, Knowledge Profiles help ensure that Program Outcomes are supported by appropriate theoretical, practical, and professional knowledge.

BAETE uses Knowledge Profiles to clearly specify the depth, breadth, and nature of knowledge required to address complex engineering problems. These profiles help identify whether a problem demands fundamental engineering knowledge, specialist expertise, practical experience, or research-based understanding, ensuring that the level of complexity is appropriate and aligned with program outcomes.

6.2 Purpose of Knowledge Profiles in OBE

Knowledge Profiles are used to:

- Classify different types of engineering knowledge
- Support Program Outcomes
- Ensure balanced curriculum design
- Guide assessment and accreditation

They help faculty understand **what kind of knowledge** is being developed through courses and activities.

Knowledge Profile Categories (K1–K8)

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BAETE defines eight Knowledge Profile categories, labeled **K1 to K8**. Each category represents a specific type of knowledge.

Description of Knowledge profiles

Attribute	Title	Description
K1	Natural Science Knowledge	A systematic, theory-based understanding of the natural sciences applicable to the discipline.
K2	Mathematics and Computational Foundations	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline.
K3	Engineering Fundamentals	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
K4	Engineering Specialist Knowledge	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
K5	Engineering Design Knowledge	Knowledge that supports engineering design in a practice area.
K6	Engineering Practice and Technology	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
K7	Comprehend (Engineering, Society, Ethics, and Sustainability)	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability.
K8	Research and Scholarly Knowledge	Engagement with selected knowledge in the research literature of the discipline.

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Mapping of POs with Knowledge Profiles

PO No.	Program Outcome Title	K1	K2	K3	K4	K5	K6	K7	K8
PO 1	Engineering Knowledge	✓	✓	✓	✓				
PO 2	Problem Analysis	✓	✓	✓	✓				
PO 3	Design / Development of Solutions					✓			
PO 4	Investigation								✓
PO 5	Modern Tool Usage						✓		
PO 6	The Engineer and Society							✓	
PO 7	Environment and Sustainability							✓	
PO 8	Ethics							✓	

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Complex Engineering Problems (CEP) and Engineering Activities (CEA)

In Outcome-Based Education (OBE), **Complex Engineering Problems (CEP)** and **Complex Engineering Activities (CEA)** play a central role in demonstrating whether students can apply their knowledge and skills to real-world engineering contexts. According to BAETE, engineering graduates must be able not only to understand theory but also to solve complex problems and perform complex activities that involve uncertainty, multiple constraints, and professional judgment. This chapter explains CEP and CEA in a practical and faculty-friendly manner, with clear examples, mapping, justification, and common pitfalls.

Understanding CEP and CEA

Complex Engineering Problem (CEP) refers to a problem that cannot be solved using routine methods alone. It typically involves incomplete or conflicting requirements, multiple

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stakeholders, and the need to apply higher-order analysis and engineering judgment.

Complex Engineering Activity (CEA) refers to tasks or activities where students design, implement, test, or evaluate engineering solutions under realistic constraints. CEA focuses more on *doing*, while CEP focuses more on *solving*.

Key Characteristics (BAETE Perspective)

A task is considered **complex** when it involves one or more of the following:

- Requires in-depth analysis and engineering judgment
- Has multiple or conflicting constraints (technical, ethical, economic, societal)
- Cannot be solved by standard textbook procedures alone
- Requires integration of knowledge from different areas
- Involves uncertainty or incomplete information

Faculty should remember that **not every assignment or lab task qualifies as CEP or CEA**. Complexity must be purposefully designed.

Program Outcomes Related to Complex Engineering Problems (CEP)

Complex Engineering Problems mainly contribute to the **PO1 to PO7**. These POs emphasize deep knowledge, analytical thinking, modeling, and research-based problem solving, which are core aspects of CEP.

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CEP attributes with descriptions.

Attribute	Description
P1: Depth of knowledge required	Cannot be resolved without in-depth engineering knowledge at the level of one or more of **K3, K4, K5, K6 or K8** which allows a fundamentals-based, first principles analytical approach.
P2: Range of conflicting requirements	Involve wide-ranging or conflicting technical, engineering and other issues.
P3: Depth of analysis required	Have no obvious solution and require abstract thinking and originality in analysis to formulate suitable models.
P4: Familiarity of issues	Involve infrequently encountered issues.
P5: Extent of applicable codes	Are outside problems encompassed by standards and codes of practice for professional engineering.
P6: Extent of stakeholder involvement and conflicting requirements	Involve diverse groups of stakeholders with widely varying needs.
P7: Interdependence	Are high-level problems including many component parts or sub-problems.

Program Outcomes Related to Complex Engineering Activities (CEA)

Complex Engineering Activities mainly contribute to the PO 10: Communication. This PO reflects professional engineering practice, teamwork, tools, and communication, which are central to CEA.

CEA attributes with descriptions.

Attribute	Description
A1: Range of resources	Involve the use of diverse resources, including people, money, equipment, materials, information, and technologies.
A2: Level of interaction	Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering, or other issues.
A3: Innovation	Involve creative use of engineering principles and research-based knowledge in novel ways.

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A4: Consequences for society and the environment	Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation.
A5: Familiarity	Can extend beyond previous experiences by applying principles-based approaches.

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Questions Commonly Asked to Students

During accreditation visits, evaluators often interact directly with students to understand their actual learning experience. These questions are not intended to test theoretical knowledge of OBE terminology, but rather to verify whether outcome-based education is genuinely practiced and visible at the student level.

Do you know the Program Outcomes of your program?

Students are expected to demonstrate general awareness that Program Outcomes describe the skills and competencies they are expected to achieve by the time of graduation. These typically include problem-solving ability, teamwork, ethical responsibility, communication skills, and technical competence.

Why This Is Asked: To assess the visibility and awareness of Outcome-Based Education among students.

Do your courses have Course Outcomes?

Students should confirm that Course Outcomes are communicated at the beginning of each course. They should

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understand that COs describe what they are expected to learn and be able to demonstrate upon completion of the course.

Why This Is Asked: To ensure that Course Outcomes are clearly shared and understood by students, rather than remaining as internal documentation.

Are assessments linked to learning outcomes?

Students should indicate that examinations, laboratory tasks, assignments, and projects are designed in alignment with the stated course outcomes.

Why This Is Asked: To confirm the practical implementation of outcome-based assessment and alignment between teaching and evaluation.

Have you worked on complex or real-world engineering problems?

Students are expected to mention experiences in laboratories, course projects, case studies, or Final Year Design Projects where they addressed complex or realistic engineering challenges.

Why This Is Asked: To verify exposure to Complex Engineering Problems (CEP) as required by accreditation standards.

Have you done team-based or project work?

Students should confirm participation in group assignments, collaborative projects, and presentations where they practiced teamwork, communication, and task management.

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Why This Is Asked**:** To evaluate development of teamwork, leadership, communication skills, and engagement in Complex Engineering Activities (CEA).

Do you receive feedback and observe improvements in courses?

Students should indicate that they receive feedback on assessments and that course delivery improves over time based on suggestions and evaluations.

Why This Is Asked: To validate the visible impact of Continuous Quality Improvement (CQI) at the student level.

How many classes did you have in your last semester, and were any makeup classes conducted? If so, how were they managed?

Students should be able to confirm that the scheduled number of classes was completed and that any missed sessions were compensated through properly arranged makeup classes with prior notice and documentation.

Why This Is Asked: To verify effective academic planning, class management, continuity of instruction, and adherence to the approved teaching schedule.

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Semester Credit Hours

- **Lecture (Theory):** *minimum 750 minutes* of formalized classroom instruction per semester **for 1 credit**
- **Laboratory:** *minimum 1500 minutes* of classroom/lab/studio/project contact per semester **for 1 credit**

How to calculate required number of classes

General formula

$$Required\ number\ of\ classes = \frac{Required\ contact\ minutes}{Minutes\ per\ class}$$

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A) Theory course (Lecture)

BAETE requirement: 1 credit = **750 minutes**

Example (if each theory class is 90 minutes)

Minimum conduct hour of Theory Courses.

Course Credit	Required Minutes	Minutes per Class	Required Classes (Minimum)
1	750	90	750/90 = **9 classes**
2	1500	90	1500/90 = **17 classes**
3	2250	90	2250/90 = **25 classes**

B) Lab course (Laboratory)

BAETE requirement: 1 credit = **1500 minutes**

Examples

Minimum conduct hour of Lab Courses.

Lab Credit	Required Minutes	Minutes per Lab Class	Required Lab Classes (Minimum)
1	1500	120	1500/120 = **13 lab classes**
1.5	2250	180	2250/180 = **13 lab classes**
2	3000	180	3000/180 = **16 lab classes**

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